THE CARTESIANITY CONDITION FOR A COMPARISON WITH THE CHAOTIC TOPOLOGY MAKES EPSILON^* AND R EPSILON_* INVERSE EQUIVALENCES ONTO THE QUASI-COHERENT OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is the author's complete argument: the fullness reduction, the choice of a K-flat representative, the Leray computation of the round trip over an arbitrary object U, the identification of the resulting tensor-up complex using quasi-coherence of M, and the appeal to condition (2) are all written out.

Editorial note. The label lemma-cartesion-plus-topology names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The comparison situation with a coarser topology tau and the functors epsilon^*, R epsilon_*. Same chapter, the section comparing the topology tau with tau' and the observations made there about epsilon^* and R epsilon_*. Supplies the functors whose equivalence is asserted and the observations used without further mention.
- Quasi-coherent objects QC(O) of D(C_tau', O). Same chapter, the definition of QC(O) by the cartesianity of the derived global sections functor. Supplies the subcategory appearing on both sides of the asserted equivalence and the condition used at the end of the proof.
- K-flat resolutions and their pullbacks. Same chapter, the lemma stating that the pullback of a K-flat complex with flat terms is again K-flat, used to see that M^bullet(U) is K-flat.. Supplies the choice of the K-flat representative M^bullet used in the computation.

Source reference. Original lemma 06YC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3508–3508. Prerequisite pointer invoked at proof line(s) 11661; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original section 073L: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6177–6177. Prerequisite pointer invoked at proof line(s) 11611; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\epsilon : (\mathcal{C}_\tau, \mathcal{O}_\tau) \to (\mathcal{C}_{\tau'}, \mathcal{O}_{\tau'})$ be as in Section 073L. Assume*

(1) *τ' is the chaotic topology on the category $\mathcal{C}$,*
(2) *for all $U \in \mathrm{Ob}(\mathcal{C})$ and all K-flat complexes of $\mathcal{O}(U)$-modules $M^\bullet$ the map*

$$M^\bullet \longrightarrow R\Gamma((\mathcal{C}/U)_\tau, (M^\bullet \otimes_{\mathcal{O}(U)} \mathcal{O}_U)^\#)$$

is a quasi-isomorphism (see proof for an explanation).

Then ϵ^* and $R\epsilon_*$ define mutually quasi-inverse equivalences between $QC(\mathcal{O})$ and the full subcategory of $D(\mathcal{C}_\tau, \mathcal{O}_\tau)$ consisting of objects K such that $R\epsilon_* K$ is in $QC(\mathcal{O})$[1]*.*

Proof. We will use the observations made in Section 073L without further mention. Since $R\epsilon_*$ is fully faithful and $\epsilon^* \circ R\epsilon_* = \mathrm{id}$, to prove the lemma it suffices to show that for M in $QC(\mathcal{O})$ we have $R\epsilon_*(\epsilon^* M) = M$. Condition (2) is exactly the condition needed to see this. Namely, we choose a K-flat complex $\mathcal{M}^\bullet$ of $\mathcal{O}$-modules with flat terms representing M. Then we see that $\epsilon^* M$ is represented by the τ-sheafification $(\mathcal{M}^\bullet)^\#$ of $\mathcal{M}^\bullet$. Let $U \in \mathrm{Ob}(\mathcal{C})$. By Leray we get

$$R\Gamma(U, R\epsilon_*(\epsilon^* M)) = R\Gamma((\mathcal{C}/U)_\tau, (\mathcal{M}^\bullet)^\#|_{\mathcal{C}/U}) = R\Gamma((\mathcal{C}/U)_\tau, (\mathcal{M}^\bullet|_{\mathcal{C}/U})^\#)$$

The last equality since sheafification commutes with restriction to $\mathcal{C}/U$. As usual, denote $\mathcal{O}_U$ the restriction of $\mathcal{O}$ to $\mathcal{C}/U$. Consider the map

$$\mathcal{M}^\bullet(U) \otimes_{\mathcal{O}(U)} \mathcal{O}_U \longrightarrow \mathcal{M}^\bullet|_{\mathcal{C}/U}$$

of complexes of $\mathcal{O}_U$-modules (in τ'-topology). By our choice of $\mathcal{M}^\bullet$ the complex $\mathcal{M}^\bullet(U)$ is a K-flat complex of $\mathcal{O}(U)$-modules; see Lemma 06YC and use that the inclusion of U into $\mathcal{C}$ defines a morphism of ringed topoi $(Sh(pt), \mathcal{O}(U)) \to (Sh(\mathcal{C}_{\tau'}), \mathcal{O})$. Since M is in $QC(\mathcal{O})$ we conclude that the displayed arrow is a quasi-isomorphism. Since sheafification is exact, we see that the same remains true after sheafification. Hence

$$R\Gamma(U, R\epsilon_*(\epsilon^* M)) = R\Gamma((\mathcal{C}/U)_\tau, (M^\bullet \otimes_{\mathcal{O}(U)} \mathcal{O}_U)^\#)$$

and assumption (2) tells us this is equal to $R\Gamma(U, M) = \mathcal{M}^\bullet(U)$ as desired. □

[1] This means that $R\Gamma(V, K) \otimes^{\mathbf{L}}_{\mathcal{O}(V)} \mathcal{O}(U) \to R\Gamma(U, K)$ is an isomorphism for all $U \to V$ in $\mathcal{C}$.